

Lowercase snake_case Migration Plan — Constitution-Compliance Phase 6

Status: PLAN — RESEARCH + AUDIT ONLY (no renames executed in this commit). **Phase:** Constitution-Compliance plan Phase 6 (per docs/plans/2026-05-15-constitution-compliance.md). **Authority:** HelixConstitution §11.4.29 (Lowercase-Snake_Case-Naming Mandate, 2026-05-15). **Companion artifacts:** scripts/audit-snake_case-references.sh (audit tool), tests/check-constitution/test_audit_snake_case_references.sh (hermetic test), docs/scripts/audit-snake_case-references.sh.md (user guide). **Classification:** project-specific (rename targets + reference inventory are Lava-specific; the rename-with-reference-update discipline is universal per §11.4.29). **Inheritance:** §11.4.29 + §11.4.17 + §11.4.18 + §11.4.20 + §11.4.26 + §11.4.27 + §11.4.32 + §6.J/§6.L (anti-bluff) + §6.W (mirror policy) + Lava's §6.AD HelixConstitution Inheritance.

0. Scope statement (no-bluff)

This document is the **RESEARCH + PLAN** deliverable for Phase 6. The actual renames are NOT executed in the commit landing this plan; they are scheduled for follow-up cycles after operator review. Per §11.4.29 itself:

"Phased execution. Comprehensive brainstorming → phase-divided plan → fine-grained tasks/subtasks with enormous detail, every change covered by every applicable test type."

The deliverables of THIS cycle are:

1. This plan document (per-submodule mapping + risk + phases + rollback).
2. scripts/audit-snake_case-references.sh — read-only audit producing a tab-separated baseline reference count.

- 3. tests/check-constitution/test_audit_snake_case_references.sh — hermetic test of the audit script.
- 4. docs/scripts/audit-snake_case-references.sh.md — script user guide per §11.4.18.
- 5. Updated Phase 6 section in docs/plans/2026-05-15-constitution-compliance.md referencing this plan + the audit script.

The rename execution is intentionally NOT in this cycle's diff. Per §6.L (Anti-Bluff Functional Reality Mandate, operator's standing order): renaming 17 submodules + 124 referencing files in a single commit without per-batch verification would be the textbook bluff this constitution forbids. The phased execution below sequences each rename batch so each one's anti-bluff evidence can be captured independently.

1. Current state baseline (captured 2026-05-16)

Captured by running the audit script (scripts/audit-snake_case-references.sh) at branch plan/2026-05-16-snake-case-migration HEAD. The full machine-readable output is in §10 of this document and reproducible via `bash scripts/audit-snake_case-references.sh`.

1.1 Top-line counts

Metric	Count
Files containing submodules/ references	124
Total submodules/ references	806
Total bare-word Submodules references (incl in paths)	829
Submodules currently registered in .gitmodules (vasic-digital + HelixDevelopment)	17
Constitution submodule (separate, see §3)	1

1.2 Per-submodule reference inventory

Submodule (current name)	Target name (snake_case)	Owner org	Refs (count)	Ref-bearing files	Rename com
Auth	auth	vasic-digital	16	9	LOW

Submodule (current name)	Target name (snake_case)	Owner org	Refs (count)	Ref- bearing files	Rename com
Cache	cache	vasic-digital	36	15	LOW
Challenges	challenges	vasic-digital	14	8	LOW
Concurrency	concurrency	vasic-digital	5	4	LOW
Config	config	vasic-digital	10	6	LOW
Containers	containers	vasic-digital	370	63	HIGH (matrix runner, emula orchestration, shaping code)
Database	database	vasic-digital	8	5	LOW
Discovery	discovery	vasic-digital	5	4	LOW
HelixQA	helix_qa (see §3.1)	HelixDevelopment	13	6	MEDIUM (HelixDevelop owned; cross- coordination)
HTTP3	http3	vasic-digital	20	7	LOW
Mdns	mdns	vasic-digital	14	7	LOW
Middleware	middleware	vasic-digital	14	6	LOW
Observability	observability	vasic-digital	29	8	LOW
RateLimiter	rate_limiter	vasic-digital	50	13	MEDIUM (hyp equivalent + c change)
Recovery	recovery	vasic-digital	12	6	LOW
Security	security	vasic-digital	15	6	LOW
Tracker-SDK	tracker_sdk	vasic-digital	83	25	HIGH (hyphen→und + composite-b Gradle includ + lava-api-go g replace direct

Total per-submodule references: 714 (close to but not equal to the 806 submodules/ total because some references count submodules/ paths that don't include a specific submodule name — e.g. submodules/ <name>/... regex matches some bare-prefix lines).

1.3 Top files by submodules/ reference count

The rename will rewrite content in these files most heavily:

File	Refs
docs/superpowers/plans/2026-05-05-anti-bluff-mandate-reinforcement-group-b.md	77
docs/superpowers/specs/2026-04-28-sp2-go-api-migration-design.md	61
docs/superpowers/plans/2026-05-05-anti-bluff-mandate-reinforcement-group-a-prime.md	54
docs/superpowers/plans/2026-04-28-sp2-go-api-migration.md	43
docs/superpowers/plans/2026-05-06-phase1-api-auth.md	41
docs/superpowers/plans/2026-05-05-pkg-vm-image-cache.md	38
CLAUDE.md	35
.gitmodules	34
scripts/check-constitution.sh	29

1.4 Distribution by top-level directory

Directory	Files affected
docs/ (including superpowers specs + plans + scripts)	41
.lava-ci-evidence/	27
lava-api-go/ (Go service incl. go.mod replace directives)	20
scripts/	16
tools/	5
tests/	5
core/ (Lava-side)	2
start.sh	1
specs/	1
settings.gradle.kts	1
CLAUDE.md	1
CHANGELOG.md	1
AGENTS.md	1
.gitmodules	1

Directory	Files affected
.githubhooks/	1

2. LANG-spec subtree exemption per §11.4.29

§11.4.29 carves out language/technology-mandated case from the snake_case rule. The exemption applies INSIDE a language-root subtree; the submodule's root directory still follows our convention.

2.1 Lava-side exempt subtrees (PascalCase or camelCase preserved)

Subtree	Reason	Exempt content
app/src/**/ *.kt	Kotlin compiler requires PascalCase class files	All *.kt source files, all package directories (already lowercase)
core/**/*.kt	Kotlin	Same as above
feature/**/ *.kt	Kotlin	Same as above
app/src/main/ res/ drawable*/, mipmap*/, etc.	Android resource directories use mandated names	drawable-*dpi, mipmap-anydpi-v26, etc.
app/src/main/ res/values*/ strings.xml (etc.)	Android resource value naming	XML resource files
buildSrc/src/ main/kotlin/ *.kt	Kotlin convention plugin file naming	*.kts convention plugins (e.g. lava.android.application.gradle.kts)
lava-api-go/ **/*.go	Go source: file names are	Confirmed Go-idiomatic

Subtree	Reason	Exempt content
	lowercase but use _ in standard Go style — already compliant	
docker/ and Dockerfile*	Docker tooling convention Dockerfile is the canonical name	Already established
Gradle wrapper: gradlew, gradlew.bat, gradle/ wrapper/	Gradle- mandated naming	Already standard
.git/, .githooks/ node_modules/ (if ever introduced)	Git- mandated dotfile names npm- mandated	Already standard N/A currently

2.2 Per-submodule LANG-spec roots

Each submodule's internal source-language layout is exempt inside its root directory. The submodule's **root directory itself** MUST be snake_case (the whole point of §11.4.29).

Submodule	Primary language	Exempt internal subtrees
Auth, Cache, Challenges, Concurrency, Config, Database, Discovery, HelixQA, HTTP3, Mdns, Middleware, Observability, RateLimiter, Recovery, Security, Tracker-SDK	Kotlin (multi- platform / Android / JVM)	**/*.kt, src/main/ kotlin/**/, src/ test/kotlin/**/, Android res/, Gradle conventions

Submodule	Primary language	Exempt internal subtrees
Containers	Go	**/*.go, cmd/ <name>/, pkg/<name>/ (already lowercase per Go convention)

2.3 Lava root-level directories that look like they need rename — actually exempt

Directory	Reason for exemption
Upstreams/	§11.4.29 specifies this directory: install_upstreams supports both Upstreams/ AND upstreams/ during migration window. Rename is OWED but the dual-support installer makes it safe to defer.
submodules/	THE rename target of Phase 6a — listed here for completeness; the rule applies.
CLAUDE.md, AGENTS.md, CHANGELOG.md, README.md, LICENSE	Convention-mandated file names (Markdown convention + LICENSE convention) — fall under "build-tooling artefacts" / common-sense exemption.
Constitution.md (if it ever exists at Lava root — currently lives in constitution/submodule)	Same as above.

3. Per-submodule target name mapping (full table)

Authoritative mapping for Phase 6b. Each rename is a separate sub-cycle.

#	Current name	Target snake_case	Owner org	Renaming rationale	S
1	Auth	auth	vasic-digital	Single-word PascalCase → lowercase	I

#	Current name	Target snake_case	Owner org	Renaming rationale	S
2	Cache	cache	vasic-digital	Same	L C f S P C f L L L a L C S P S C S t L L
3	Challenges	challenges	vasic-digital	Same	
4	Concurrency	concurrency	vasic-digital	Same	
5	Config	config	vasic-digital	Same	L L L a L C S P S C S t L L
6	Containers	containers	vasic-digital	Same	
7	Database	database	vasic-digital	Same	
8	Discovery	discovery	vasic-digital	Same	
9	HelixQA	helix_qa	HelixDevelopment	CamelCase compound word → snake_case with underscore. See §3.1 for HelixDevelopment-org considerations.	C
10	HTTP3	http3	vasic-digital	Acronym + digit: HTTP3 → http3 (no underscore between letters and digits — http_3 would arguably be more idiomatic snake_case but http3 matches the canonical protocol name)	L C - a
11	Mdns	mdns	vasic-digital	Single-word lowercased acronym	L
12	Middleware	middleware	vasic-digital	Same	L

#	Current name	Target snake_case	Owner org	Renaming rationale	
13	Observability	observability	vasic-digital	Same	
14	RateLimiter	rate_limiter	vasic-digital	CamelCase compound → snake_case	
15	Recovery	recovery	vasic-digital	Same	
16	Security	security	vasic-digital	Same	
17	Tracker-SDK	tracker_sdk	vasic-digital	Hyphen + acronym → underscore + lowercase. HIGH-RISK — 83 refs across 25 files including settings.gradle.kts includeBuild("submodules/tracker_sdk") AND lava-api-go/go.mod replace directives.	

3.1 HelixQA / HelixDevelopment-org consideration

HelixQA is owned by HelixDevelopment, not vasic-digital. §11.4.29 applies universally — all owned-by-us submodules (any org in the operator's list) must follow snake_case. HelixDevelopment IS on the owned-org list (per §11.4.28's verbatim list: vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory).

Decision: rename to helix_qa. The exemption clause "third-party submodules keep upstream names" does NOT apply because HelixDevelopment is OUR org. The rename is binding.

Upstream rename consideration. The submodule lives at git@github.com:HelixDevelopment/HelixQA.git upstream. §11.4.29's reach is the local checkout path — the upstream repo name itself ALSO follows §11.4.29 universally, but renaming an upstream GitHub/GitLab repo is a separate, coordinated operation:

- The .gitmodules path field is what Lava controls — we set this to submodules/helix_qa.
- The .gitmodules url field points at the upstream (HelixDevelopment/HelixQA). The upstream itself remains at its existing URL until a separate cross-org coordination renames the upstream repo too.
- After the upstream is renamed (e.g. to HelixDevelopment/helix_qa or HelixDevelopment/helix-qa per GitHub naming convention), we

update `.gitmodules` URLs and migrate. Until then, the local path is `snake_case` while the upstream URL remains `CamelCase` — this is the §11.4.29-conformant transitional state (rule applies to "directories under the parent project's working tree"; the `url` field is a network identifier, not a working-tree directory).

Equivalent for Tracker-SDK. Same logic: upstream is `vasic-digital/Tracker-SDK`. Local path becomes `submodules/tracker_sdk`. Upstream URL update is a follow-up coordination cycle (`vasic-digital` is OUR org, so we own the timing).

Equivalent for every other vasic-digital submodule. Same pattern — local path is `canonical-snake_case`; upstream URL update is per-org follow-up.

3.2 Constitution submodule special case (already mounted at `constitution/`)

The `HelixDevelopment` `constitution` submodule is mounted at `./constitution/` (lowercase, no underscore — single word). It is ALREADY `snake_case`-compliant. No rename needed. The §11.4.35 `canonical-root` inheritance pointer in `Lava's CLAUDE.md + AGENTS.md` continues to reference `constitution/CLAUDE.md`.

4. Phased execution plan

Per §11.4.29 + §11.4.20 (subagent delegation): each phase below is a separate cycle (separate commit, separate falsifiability rehearsal, separate sweep run). Phases run sequentially because each builds on the previous (e.g. Phase 6b uses the already-renamed `submodules/` path from Phase 6a).

Phase 6.0 — Plan + audit tool (THIS commit)

Status: SHIPPING IN THIS COMMIT.

Deliverables:

- This plan document.
- `scripts/audit-snake_case-references.sh` (read-only audit tool).
- `tests/check-constitution/test_audit_snake_case_references.sh` (hermetic test).
- `docs/scripts/audit-snake_case-references.sh.md` (companion doc).

- Updated reference in docs/plans/2026-05-15-constitution-compliance.md Phase 6 section.

Risk: ZERO — read-only. No renames. Audit script + hermetic test + plan only.

Exit criteria:

- Audit script runs successfully on real tree.
 - Hermetic test passes.
 - `scripts/verify-all-constitution-rules.sh --strict` reports unchanged baseline (no new gate failures).
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Phase 6a — `submodules/` → `submodules/` (the directory itself)

Scope: Single rename — the top-level Submodules directory. Cascading updates to every reference path.

Pre-rename audit:

1. Run `bash scripts/audit-snake_case-references.sh > before.tsv`.
2. Hardlinked `.git` backup per §9.1: `cp -al .git ../.git-backup-pre-snake-case-6a-$(date -u +%Y%m%dT%H%M%S)/repo.git.mirror`.

Rename steps:

1. `git mv Submodules submodules` (or equivalent: `git mv` each child entry — Git's `git mv` handles renamed-directory tracking automatically).
2. Update `.gitmodules`: every path = `submodules/<X>` → path = `submodules/<X>` (17 entries).
3. Update `.git/config submodule.*.path` mirrored from `.gitmodules`. (Usually done by `git submodule sync`.)
4. Run `git submodule sync + git submodule update --init` to refresh `.git/modules` pointers.
5. Update every script/code reference:
 - `scripts/*.sh` — `submodules/` → `submodules/` (16 files).
 - `lava-api-go/go.mod` — `../submodules/` → `../submodules/` (replace directives).
 - `settings.gradle.kts` — `includeBuild("submodules/tracker_sdk")` → `includeBuild("ssubmodules/tracker_sdk")`.
 - `start.sh`, `tools/lava-containers/...`, `tests/...`.
 - `CLAUDE.md` (35 refs), `AGENTS.md`, `CHANGELOG.md`, `.github/hooks/pre-push`.
 - `docs/**/*.md` (41 files — including superpowers specs/plans).

- `.lava-ci-evidence/**/*.*` (27 files — these are historical attestations; the rename in evidence files is for consistency, not for correctness).
- 6. Verify no remaining submodules/ references via audit: `bash scripts/audit-snake_case-references.sh > after.tsv && diff before.tsv after.tsv`.

Post-rename validation:

1. `./gradlew :app:assembleDebug` — verifies Gradle composite-build picks up the renamed path.
2. `cd lava-api-go && make build` — verifies Go module replace-directives work.
3. `bash scripts/run-emulator-tests.sh --help (smoke)` — verifies matrix runner glue.
4. `bash scripts/verify-all-constitution-rules.sh --strict` — full sweep.
5. `bash tests/check-constitution/test_audit_snake_case_references.sh` — hermetic.

Falsifiability rehearsal:

- Deliberately omit ONE rename (e.g. leave `scripts/check-constitution.sh` referencing submodules/). Run audit — confirm it surfaces the residual reference. Restore + verify clean.

Rollback:

- `git checkout HEAD -- .` (working tree).
- `git reset --hard HEAD~1` (last commit).
- Restore from `.git-backup-pre-snake-case-6a-...` if git itself is somehow corrupted.

Estimated effort: 1 focused session. Commit message MUST cite the falsifiability rehearsal per §6.N.

Operator approval required: YES — high blast radius. Operator MUST sign off on the rename batch before push to mirrors.

Phase 6b — Per-submodule `snake_case` renames (17 sub-cycles)

Strategy: ONE commit per submodule. Each commit:

1. Renames the directory.

2. Updates `.gitmodules` [submodule "..."] heading + path field.
3. Updates every reference in tracked files.
4. Runs the audit + post-rename validation.
5. Captures a per-cycle falsifiability rehearsal.

Ordering — risk-ascending (small blast radius first to build muscle):

Order	Submodule	Refs	Risk
1	Discovery	5	LOW
2	Concurrency	5	LOW
3	Database	8	LOW
4	Config	10	LOW
5	Recovery	12	LOW
6	HelixQA → helix_qa	13	MEDIUM (cross-org)
7	Challenges	14	LOW
8	Mdns	14	LOW
9	Middleware	14	LOW
10	Security	15	LOW
11	Auth	16	LOW
12	HTTP3 → http3	20	LOW
13	Observability	29	LOW
14	Cache	36	LOW
15	RateLimiter → rate_limiter	50	MEDIUM (compound name)
16	Tracker-SDK → tracker_sdk	83	HIGH (Gradle includeBuild + Go go.mod)
17	Containers → containers	370	HIGHEST (matrix runner + gate-shaping code)

Per-cycle pre-rename audit:

- `bash scripts/audit-snake_case-references.sh > before-<name>.tsv.`
- `cp -al .git ../.git-backup-pre-rename-<name>-${date -u +%Y%m%dT%H%M%S}/repo.git.mirror.`

Per-cycle rename steps:

1. `git mv submodules/<OldName> submodules/<new_name>.`

2. Update .gitmodules:

- [submodule "submodules/<OldName>"] → [submodule "submodules/<new_name>"].
- path = submodules/<OldName> → path = submodules/<new_name>.

3. Update every reference in:

- scripts/*.sh (use `git grep -l "<OldName>" | xargs sed -i 's|submodules/<OldName>|submodules/<new_name>|g'`).
- docs/**/*.md.
- CLAUDE.md, AGENTS.md, CHANGELOG.md.
- lava-api-go/go.mod (if this submodule has Go replace directives).
- settings.gradle.kts (if Gradle includeBuild references it).
- .lava-ci-evidence/**/*.md.

4. Run `git submodule sync` to refresh git's internal pointers.

5. Verify: `bash scripts/audit-snake_case-references.sh > after-<name>.tsv && diff before-<name>.tsv after-<name>.tsv`.

Per-cycle validation:

- `./gradlew :app:assembleDebug` (if this submodule is in Gradle build).
- `cd lava-api-go && make build` (if this submodule is in Go module).
- `bash scripts/verify-all-constitution-rules.sh --strict`.

Per-cycle falsifiability rehearsal:

- Leave ONE reference unrenamed (e.g. in a script). Run audit — confirm it flags the residual. Restore + verify clean.

Per-cycle commit message MUST contain:

- Bluff-Audit: block per §6.N + Seventh Law clause 1.
- Classification: line per §11.4.17 (project-specific — the rename target IS Lava's).
- Citation of falsifiability rehearsal.

Per-cycle push: to both GitHub + GitLab mirrors per §6.W.

Estimated effort: 17 sessions (one per submodule). Some pairs may bundle (small + small) at operator's discretion; HIGH-RISK renames (Containers, Tracker-SDK, RateLimiter) MUST be their own sessions.

Phase 6c — File-level renames (Lava-side)

Scope: Tracked files at Lava root or under non-LANG-spec subtrees that violate snake_case. Per §11.4.29 the rule applies to project files too.

Audit step (NEW script needed in this phase):

- scripts/audit-non-snake-case-files.sh — walks tracked files, lists every non-snake_case name not in the §2 exemption set.

Expected findings (preview):

- docs/INCIDENT_2026-04-28-HOST-POWEROFF.md — already snake_case in content but the INCIDENT_ prefix is ALL-CAPS. Decision: keep as-is per "tradition" (legacy filename established by operator; rename doc-only files only when low-risk; the rule's spirit is operator-ergonomics).
- docs/superpowers/specs/*-design.md — already snake_case.
- docs/superpowers/plans/*.md — already snake_case.
- tools/lava-containers/ — already snake_case.

Anticipated decisions:

- LANG-spec subtrees: exempted (see §2).
- Markdown docs at root: keep PascalCase / UPPERCASE per Markdown ecosystem convention (README.md, CHANGELOG.md, LICENSE, CLAUDE.md, AGENTS.md, INCIDENT_*.md).
- New docs (Phase 6c onwards) MUST be snake_case from inception.

Estimated effort: 1 session for the audit + decisions. Actual renames may be ZERO if all current files are either snake_case or in an exemption set.

Phase 6d — Upstreams/ → upstreams/ (Lava side)

Scope: Rename the Lava-side Upstreams/ directory per §11.4.29's explicit transition clause.

Pre-requisites:

- Confirm install_upstreams (host-system utility from constitution submodule) supports BOTH Upstreams/ and upstreams/ directory layouts. The constitution mandates this transitional dual-support — verify before renaming.

- Constitution submodule's `install_upstreams.sh` is already on operator's PATH.

Rename steps:

1. `git mv Upstreams upstreams`.
2. Update any script that hard-codes `Upstreams/` (currently 0 from `grep`; the `install_upstreams` utility scans both).
3. Run `install_upstreams` from repo root — verify it picks up the new lowercase directory.
4. Update `CLAUDE.md` reference (`./Upstreams/GitHub.sh` → `./upstreams/GitHub.sh`).
5. Update `CHANGELOG.md` if mentioned.

Validation:

- `git remote -v` — verify both GitHub + GitLab remotes still configured correctly.
- Push to both mirrors — verify multi-upstream push works.

Estimated effort: Half a session.

Phase 6e — Mechanical gate

Scope: `scripts/check-snake-case-naming.sh` enforcement gate + companion hermetic test, wired into `scripts/verify-all-constitution-rules.sh`.

Gate logic:

- Walk all tracked files + directories.
- For each name, check it matches `^[a-z0-9_]+(\.[a-z0-9_]+)*$` OR is on the LANG-spec exemption list (`*.kt`, `*.kts`, `*.gradle.kts`, `*.go`, `*.xml`, `*.json`, `*.yaml`, `*.yml`, `*.md` for markdown, Android `drawable-*/mipmap-*/etc.`, Gradle wrapper, dotfiles).
- Reject on violation with `"non-snake_case name: "`.

Modes:

- `--strict` (default): exit 1 on violation.
- `--advisory`: exit 0; print warnings.

Companion hermetic test:

- Plant a `BadName.txt` synthetic file — confirm gate FAILs.

- Plant a `MainActivity.kt` synthetic file — confirm gate PASSes (LANG-spec exemption).
- Confirm clean Lava tree PASSes.

Falsifiability rehearsal:

- Rename a `snake_case` file back to `PascalCase`. Run gate — confirm FAIL with the offender named.

Estimated effort: 1 session.

Phase 6f — Upstream coordination (cross-org rename of repos themselves)

Scope: Optionally rename the upstream Git repos (`github.com/vasic-digital/Tracker-SDK` → `github.com/vasic-digital/tracker_sdk` or `tracker-sdk`; `github.com/HelixDevelopment/HelixQA` → `helix_qa` / `helix-qa`).

Considerations:

- GitHub convention is kebab-case (hyphens). Renaming to `tracker-sdk` would be GitHub-idiomatic; to `tracker_sdk` would be §11.4.29-idiomatic. §11.4.29 silent on the hyphen-vs-underscore for upstream repo names — likely we keep upstream URL as-is and only the LOCAL submodule path is renamed.
- Renaming an upstream Git repo breaks every clone URL anywhere. GitHub auto-redirects old URLs but only for some operations. Recommendation: do NOT rename upstream repos unless operator explicitly authorizes; the local-path rename per §11.4.29 is what's mandatory.
- If operator authorizes: coordinate `per-org.gh repo rename vasic-digital/Tracker-SDK tracker_sdk` (or use the GitHub UI). Verify GitLab mirror name updates. Update all consuming projects' `.gitmodules url` fields.

Decision (pending operator): Defer. The §11.4.29 local-path rename is what the constitution mandates; the upstream repo name is a network identifier, not a working-tree directory.

Estimated effort: 0-1 session (depends on operator decision).

5. Rollback plan

Every rename batch has a hardlinked `.git` backup per §9.1 (HelixConstitution + Lava's existing host-stability discipline). The backup costs near-zero disk (hardlinks).

5.1 Rollback for single rename batch (during the session)

1. `git reset --hard HEAD~1` if the rename was just committed.
2. `git checkout HEAD -- .` to discard uncommitted working-tree changes.
3. If git's internal state is somehow corrupt: `rm -rf .git && cp -al ../.git-backup-pre-rename-<name>-<TS>/repo.git.mirror .git.`

5.2 Rollback after push (more involved)

1. Operator-approved force-push of the inverse commit (per §6.T.3 — force-push requires explicit per-operation approval).
2. Re-clone consuming projects if their submodule pin points at the abandoned commit.
3. Document in `.lava-ci-evidence/sixth-law-incidents/<date>-snake_case-rename-rollback.json` per §6.M.

5.3 Partial rollback (some renames stick, some revert)

Use case: Phase 6a + 6b/1-3 succeed; 6b/4 breaks something. Revert ONLY 6b/4:

1. `git revert <SHA-of-6b/4>.`
2. Push the revert commit.
3. Submodule 4's local path returns to the partially-renamed state (still capital `submodules/<OldName>` if 6b/4 was the FIRST per-submodule rename after 6a — meaning the directory is `submodules/<OldName>` after 6a but `<OldName>` was not lowered yet).

5.4 What MUST NOT be done during rollback

- Force-push without operator authorization per §6.T.3.
 - Skip pre-push hooks with `--no-verify` per §6.T.3.
 - Rewrite history via `git filter-branch` / `git filter-repo` without §9.1 backup + operator approval.
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6. Risk analysis

6.1 Risk: Gradle composite-build path break

Trigger: Phase 6a OR Phase 6b/Tracker-SDK rename without updating `settings.gradle.kts` `includeBuild("submodules/tracker_sdk")`.

Symptom: `./gradlew :app:assembleDebug` fails with "Project 'submodules/tracker_sdk' not found" or analogous error.

Mitigation: `settings.gradle.kts` IS in the audit script's reference count (1 ref). The per-cycle validation step (`./gradlew :app:assembleDebug`) catches this immediately.

6.2 Risk: Go module replace-directive path break

Trigger: Phase 6a OR Phase 6b/ without updating `lava-api-go/go.mod` `replace` directives.

Symptom: `cd lava-api-go && go build` fails with "directory ../submodules/X does not exist".

Mitigation: `lava-api-go/go.mod` IS in the audit script's reference count (15 refs in `go.mod` alone). The per-cycle validation step (`cd lava-api-go && make build`) catches this immediately.

6.3 Risk: Constitution checker reference break

Trigger: Phase 6a OR Phase 6b/ without updating `scripts/check-constitution.sh` (29 refs).

Symptom: Pre-push hook fails with stale path errors; OR pre-push hook silently passes while the rule it claims to enforce is no longer checked.

Mitigation: `scripts/check-constitution.sh` has THE HIGHEST per-script ref count. Per-cycle validation MUST run `bash scripts/verify-all-constitution-rules.sh --strict` — any silent pass on a stale path surfaces as a sweep regression.

6.4 Risk: Submodule init/sync gets confused

Trigger: Renaming `submodules/<X>` without `git submodule sync`.

Symptom: `.git/modules/submodules/X/` (cached metadata) vs. `.git/modules/submodules/x/` (post-rename expectation) divergence; `git submodule update --init` fails with "fatal: repository '.../X' does not exist".

Mitigation: Per-cycle step 4 explicitly runs `git submodule sync + git submodule update --init`. If still confused: `rm -rf .git/modules/<stale-path>` (safe — re-fetched on next update `--init`).

6.5 Risk: HelixDevelopment-owned submodule cross-org coordination

Trigger: Phase 6b/HelixQA — local path becomes `submodules/helix_qa` while upstream URL still `HelixDevelopment/HelixQA` (PascalCase).

Symptom: None expected. The `path` and `url` fields in `.gitmodules` are independent. §11.4.29 governs working-tree paths, not network identifiers.

Mitigation: Document the transitional state in the commit message. If operator later authorizes the upstream rename, run Phase 6f in coordination with HelixDevelopment.

6.6 Risk: HIGH-blast-radius rename batches (Containers + Tracker-SDK)

Trigger: Phase 6b/Containers (370 refs) or Phase 6b/Tracker-SDK (83 refs).

Symptom: Higher chance of missing a reference; higher diff-volume in the commit; harder for reviewer to verify the rename is complete.

Mitigation:

- Run the audit script BOTH before AND after; the `diff` MUST show every old-name reference removed (the audit's tab-separated output makes this easy).
- For Containers: separate the rename of `submodules/containers/` itself from updates to scripts that reference it. Could be split into two sub-commits within the cycle (rename + path-update).
- For Tracker-SDK: explicitly test the Gradle composite-build path with `./gradlew :app:assembleDebug` AND Go module path with `cd lava-api-go && go build` BEFORE pushing.

6.7 Risk: Reference drift between rename batches

Trigger: Phase 6b/N renames submodules/X, but Phase 6b/N+1 commit lands with stale references to submodules/X (the renamed-old-name) somewhere.

Symptom: §11.4.29 violation per "reference drift after a rename is a §11.4.29 violation of equal severity to the rename itself".

Mitigation: The audit script (run as pre-push gate after each rename) catches this immediately. The mechanical gate from Phase 6e (when wired) will reject pushes with stale references automatically.

6.8 Risk: Operator-unrelated breakage during the migration window

Trigger: Operator's local working tree gets out of sync because of a half-completed pull (Phase 6a landed remotely; operator's checkout is pre-rename).

Symptom: `git status` shows submodules/ and submodules/ simultaneously (case-insensitive filesystems on macOS could confuse this further).

Mitigation: Operator MUST run `git status + git submodule status` after every pull during the migration window. Document the rename batches in `CHANGELOG.md` so each rename has a discoverable record. macOS case-insensitive filesystems may need `git rm --cached Submodules + git submodule add` cycles in extreme cases.

7. Anti-bluff guarantees (per §11.4.29 test-coverage clause + §6.J / §6.L)

§11.4.29 explicitly mandates:

"Test coverage of renames. Every batch of renames MUST ship with: (i) a regression test that verifies every reference to the renamed entity now resolves to the new name (no stale references left); (ii) a full CONST-050(B) test-type matrix run against the post-rename tree; (iii) anti-bluff (CONST-035) wire-evidence captured during the runtime verification."

Per-phase coverage:

7.1 Regression test: zero stale references

The audit script (`scripts/audit-snake_case-references.sh`) IS the regression detector. Per-phase exit criterion: `audit_after - audit_before == 0` for the renamed-from name. The hermetic test verifies the script's correctness on planted fixtures.

7.2 Full test-type matrix on post-rename tree

After each Phase 6b sub-cycle:

- Unit tests: `./gradlew test --rerun-tasks (Lava) + cd lava-api-go && make test (Go)`.
- Integration: `./gradlew :app:connectedDebugAndroidTest` (Compose UI Challenge Tests on emulator — gated by Containers submodule per §6.X / §6.AE).
- Constitution sweep: `bash scripts/verify-all-constitution-rules.sh --strict`.
- Pre-push hook: implicit on push.

7.3 Anti-bluff wire evidence

Per-phase JSON evidence file at `.lava-ci-evidence/snake_case-migration/<phase>-<UTC-timestamp>.json` capturing:

- Before/after audit TSV output diff.
- Build verification command stdout/stderr.
- Sweep result.
- Falsifiability rehearsal output.

7.4 Falsifiability rehearsal (§6.J clause 2 + §6.N clauses 2-3)

Per-cycle: leave ONE reference unrenamed (e.g. inside an evidence file or a docs/superpowers/plans/ entry); run audit — confirm it flags the residual reference; restore. The Bluff-Audit: stamp in the commit body records the residual + the observed audit failure message.

8. Composition with other constitutional rules

§11.4.29 composes with:

- **§1 (four-layer floor)** — every rename batch runs pre-build + post-build + runtime + meta-test paired mutation.
- **§1.1 (paired mutation)** — falsifiability rehearsal per cycle (§7.4 above).
- **§11.4.12 (auto-generated docs sync)** — if any auto-generated doc references the old name, regenerate in the same commit.
- **§11.4.17 (universal-vs-project classification)** — rename batch commits MUST carry `Classification: line` (project-specific — target names are Lava-specific).
- **§11.4.18 (script-doc sync)** — every renamed `scripts/*.sh` requires same-commit update to `docs/scripts/*.md`.
- **§11.4.20 (subagent delegation)** — Phase 6b can run multiple sub-cycles in parallel via subagents for non-overlapping submodules.
- **§11.4.25 (coverage ledger)** — every renamed path appears in the coverage ledger when Phase 7 ships.
- **§11.4.26 (constitution-submodule update workflow)** — if a rename triggers any constitution update, follow the 7-step pipeline.
- **§11.4.27 (no-fakes-beyond-unit-tests + 100%-test-type-coverage)** — every rename batch covered by every applicable test type.
- **§11.4.28 (Submodules-As-Equal-Codebase)** — rename batches treat owned submodules as equal codebase; each gets engineering attention.
- **§11.4.32 (post-pull validation sweep)** — operator MUST run `scripts/verify-all-constitution-rules.sh` after pulling any rename batch.

Lava-specific composition:

- **§6.J / §6.L (anti-bluff)** — every rename batch's commit body carries a Bluff-Audit: stamp recording the falsifiability rehearsal.
 - **§6.W (mirror policy)** — push to both GitHub + GitLab mirrors per cycle.
 - **§6.S (CONTINUATION maintenance)** — `docs/CONTINUATION.md` updated in the same commit as any rename batch.
 - **§6.AD-debt (HelixConstitution gate-wiring)** — the rename mechanical gate (Phase 6e) closes part of this debt.
-

9. Subagent delegation plan (per §11.4.20)

§11.4.20 (subagent-driven-by-default mandate): when scope is multi-step + parallelizable, the primary agent **MUST** delegate.

The 17 per-submodule renames in Phase 6b are textbook parallelizable IF they touch non-overlapping files. They don't always — `CLAUDE.md` is touched by EVERY rename batch — so parallel delegation requires coordination.

Recommended delegation pattern:

- **Phase 6a:** primary agent (high-coordination, touches everything).
 - **Phase 6b/1-5 (LOW-RISK, ≤12 refs each):** primary agent or 1 subagent per cycle, sequential.
 - **Phase 6b/6-13 (LOW-MEDIUM, 13-30 refs each):** primary agent OR up to 2 subagents in parallel IF they partition disjoint files (one renames submodules/<X> and updates docs/superpowers/plans/<P>.md; the other renames submodules/<Y> and updates docs/superpowers/plans/<Q>.md).
 - **Phase 6b/14-17 (HIGH-RISK, ≥36 refs each):** primary agent, sequential. NO parallel delegation. These are high-coordination renames where reviewer attention matters.
 - **Phase 6c, 6d, 6e:** primary agent.
 - **Phase 6f:** OPERATOR decision required (cross-org upstream rename).
-

10. Audit script baseline output (captured 2026-05-16)

Reproducible via `bash scripts/audit-snake_case-references.sh`. The script's tab-separated output is the authoritative source for §1.

```
# Captured at branch plan/2026-05-16-snake-case-migration HEAD
NAME      REFS    FILES
Auth      16      9
Cache     36      15
Challenges 14      8
Concurrency 5       4
Config    10      6
Containers 370     63
Database  8       5
Discovery 5       4
HelixQA   13      6
```

HTTP3	20	7
Mdns	14	7
Middleware	14	6
Observability	29	8
RateLimiter	50	13
Recovery	12	6
Security	15	6
Tracker-SDK	83	25
TOTAL_Submodules	806	124

11. Open questions for operator review

- 1. Phase 6f (upstream repo rename) — proceed or defer?** Current recommendation: defer. §11.4.29 governs local working-tree paths; upstream repo names are network identifiers.
- 2. HelixQA local-path target: helix_qa or helixqa?** Current recommendation: helix_qa (snake_case requires word separator; Helix and QA are clearly two words). Alternative: helixqa (single token, matches mdns / http3 precedent for short compound names). Decision-axis: word-boundary clarity.
- 3. HTTP3 local-path target: http3 or http_3?** Current recommendation: http3 (matches the protocol's canonical name; digits inside a single token, no separator needed). Alternative: http_3 (pedantic snake_case). Decision-axis: ergonomics vs. pedantry. The mdns / nrmclub / gutenber / rutracker neighbors all use single-token names without internal separators.
- 4. RateLimiter local-path target: rate_limiter or ratelimiter?** Current recommendation: rate_limiter (compound noun with clear word boundary). Alternative: ratelimiter (matches Go pkg/ratelimiter convention common in Go code). Decision-axis: snake_case fidelity vs. Go-idiomatic single-token.
- 5. Container CLI subcommand paths: pkg/emulator etc. — already snake_case.** No action needed for inside-Containers paths; Go-native.
- 6. submodules/containers/cmd/distributed-build/ (and siblings with hyphens)** — Go convention permits hyphens in cmd/<name>/ directory names. §11.4.29 LANG-spec exemption applies. Verify with the operator that Go-tooling-convention hyphens are EXEMPT from the rule. Current recommendation: EXEMPT.

7. **Order-of-operations dependency:** Phase 6a MUST land before Phase 6b. Phase 6b/17 (Containers — 370 refs) is the highest-risk rename and SHOULD land LAST in Phase 6b. Phase 6c-f follow. Confirm operator agrees with the ordering.
8. **Phase 6f-style upstream rename for Tracker-SDK (vasic-digital/Tracker-SDK → vasic-digital/tracker_sdk)?** Same defer recommendation as HelixQA.
-

12. Sign-off checklist for execution

Before Phase 6a executes (separate commit from this plan):

- ☐ Operator reviews this plan.
 - ☐ Operator approves §11 open questions (or sends amendments).
 - ☐ Operator approves the per-cycle ordering in §4.
 - ☐ Operator reads §5 rollback plan.
 - ☐ Operator reads §6 risk analysis.
 - ☐ Linux x86_64 gate host available for Phase 6a + 6b/17 (Containers rename) so emulator-matrix attestation can run if §6.X-debt closes in the interim.
 - ☐ .git-backup-pre-snake-case-6a-<TS>/ directory plan confirmed (disk space, location, retention).
 - ☐ CONTINUATION.md updated to reflect the migration window opening.
-

End of plan. This document is the authoritative reference for the entire snake_case migration program. Per §6.S, it MUST be kept in sync with reality as phases execute; per-phase commits MUST update docs/CONTINUATION.md AND any references herein that have shifted.